Hyunkyu Kang

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RESEARCH INTERESTS

My research interests lie in human–scene interaction, affordance-aware motion generation, Robotics, and embodied AI. I study how language, spatial relationships, and interaction history can guide context-appropriate behavior, with an emphasis on reliable data pipelines and reproducible evaluation.

EDUCATION

Chung-Ang University **March 2021 - February 2027(Expected)**

Bachelor of School of Art and Technology

Bachelor of Science in Cyber Security (Convergence Major)

RESEARCH EXPERIENCE

Chung-Ang University **Jul 2026 – Current**

Coexistent Reality and Intelligent Agents Lab | Advisor: Prof. Taeil Jin

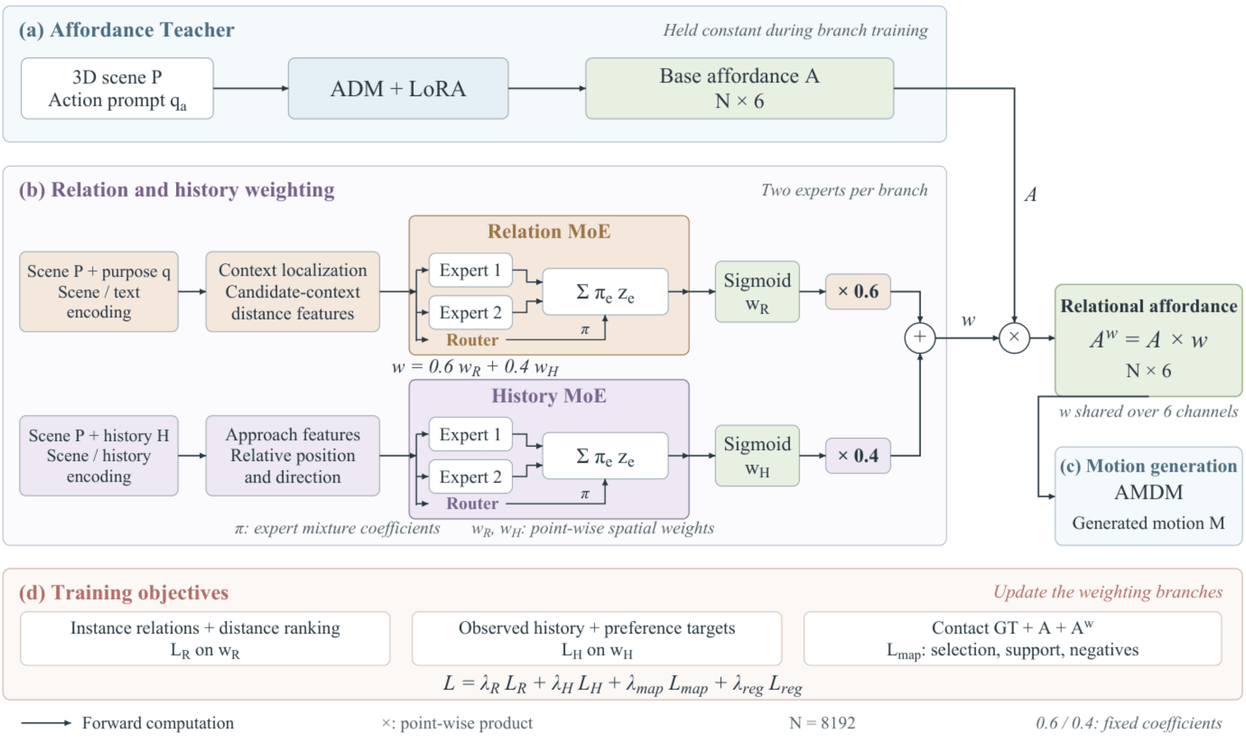
- Worked on language-based 3D scene understanding and purpose- and history-conditioned affordance maps.
- Designed a scene-point/body-part representation and implemented motion preprocessing, coordinate validation, and teacher–student training and evaluation components.

REPRESENTATIVE RESEARCH

Purpose- and History-Conditioned Affordance-Aware Motion Generation

Ongoing research | *Code*

- Designed an affordance representation over 8,192 scene points and six body parts, combining purpose, scene geometry, and interaction history.
- Implemented teacher adaptation, contact supervision, point-alignment checks, and checkpoint auditing within an ADM/LoRA and mixture-of-experts research workflow.
- Established strict multi-object evaluation gates and documented checkpoint limitations; end-to-end performance remains under evaluation.



Proposed teacher–student pipeline for purpose- and history-conditioned affordances. End-to-end performance remains under evaluation.

RESEARCH AND PROJECTS

Video2Unity: Transformer-Based Motion Refinement

AI & ML course project, Fall 2025 | Code

- Developed a video-to-animation pipeline using BlazePose 3D landmarks and a TensorFlow/Keras Transformer encoder to refine 33-joint poses from 30-frame sequences.
- Applied root-centered normalization, denoising training, Savitzky-Golay smoothing, and floor alignment to address temporal jitter and grounding artifacts.
- Integrated refined motion into a Unity dance performance with motion interpolation, scripted camera tracking, and lighting; evaluated motion smoothness and foot-contact behavior on a held-out video.

Video-to-Robot Motion Transfer

Simulation prototype | Code

- Connected reference-video interpretation to structured motion specifications, JAX reward functions, and Dial-MPC control for a Unitree Go2 model in MuJoCo/MJX.
- Worked on startup holds, velocity ramps, warm starts, and joint-limit safeguards. Validation is simulation-based, not a physical-robot deployment.

Korean Hate-Speech Robustness under Text Obfuscation

NLP evaluation | Code

- Evaluated a balanced 500-comment K-MHaS subset across five variants (2,500 predictions per system), comparing character-based, KoELECTRA, and Qwen-based pipelines.
- Measured normalization and calibration trade-offs; normalization + character TF-IDF MLP achieved 0.772 worst-condition balanced accuracy on the evaluated benchmark.

TECHNICAL EXPERIENCE

Programming and ML: Python, PyTorch, NumPy, scikit-learn, Hugging Face Transformers

Motion and simulation: HumanML263, Unity/YBot motion processing, MuJoCo/MJX, JAX

Research workflows: Git, Linux, LoRA adaptation, experiment auditing, visualization